

Multi-Agent Ambulance Dispatch & Hospital Allocation

Project Review 1: PEAS Formulation, Environment Analysis & Algorithmic Search Strategy

MEMBER 1

CB.SC.U4CSE23547

Slides 2 & 3

Architecture & Dispatch/Ambulance
PEAS (3M)

MEMBER 2

CB.SC.U4CSE23538

Slides 4 & 5

Hospital PEAS, ESI & 7-Dimensions
(3M)

MEMBER 3

CB.SC.U4CSE23535

Slides 6 & 7

Agent Typology & Contract Net
Protocol (3M)

MEMBER 4

[Member 4 Roll No.]

Slides 8, 9, 10

A* Search, Gale-Shapley & Roadmap
(4M)

REVIEW 1 EVALUATION MATRIX • 10 MARKS DISTRIBUTION

3 Marks

PEAS Formulation

3 Marks

Environment & Agent Analysis

3 Marks

Algorithmic Search & Modeling

1 Mark

Q&A & Presentation Mechanics

Problem Statement & Multi-Agent Architecture

System Overview



Critical Bottlenecks in Modern EMS

- **Single-Point Computational Bottlenecks:** Central dispatchers face exponential queue delays during mass-casualty surges and severe weather events.
- **Ambulance Offload Delay (AOD):** Patients spend deadly hours stranded in idling ambulances outside saturated ERs due to lack of pre-arrival bed coordination.
- **Information Asymmetry:** Dispatch centers lack real-time ER surgical readiness; hospitals lack real-time inbound patient vital telemetry.
- **Decentralized MAS Solution:** Decomposes the global assignment into localized, autonomous bidding agents and bilateral hospital bed reservation protocols.



Multi-Agent System (MAS) Execution Flowchart

1. CALLER / INCIDENT INTAKE

Caller GPS coordinates & initial triage symptoms (ESI 1-5).



2. DISPATCH COORDINATOR AGENT

Broadcasts Call for Proposals (CFP) to all active patrol units.



3. AMBULANCE FLEET AGENTS

Computes dynamic Time-Dependent A* bids and submits proposals.



4. HOSPITAL ER ALLOCATION AGENTS

Bilateral auction locks trauma/ICU bed prior to ambulance arrival.

PEAS Formulation: Dispatcher & Ambulance Agents

Rubric: PEAS (3 Marks)

Agent Entity	Performance Measure (P)	Environment (E)	Actuators (A)	Sensors (S)
Dispatch Coordinator Agent (System Orchestrator)	• Mean response time (< 8 min) • ESI triage adherence (%) • Jain's geographic fairness index • Coverage void minimization	• City road network graph G=(V,E) • Stochastic 911 incoming call stream • Active ambulance fleet states • Regional hospital cluster status	• CFP broadcast transmitter • Contract award signal • Fleet re-allocation directive • Priority escalation broadcast	• CAD 911 incoming call feed • Caller geocoded GIS location • Ambulance GPS beacons • Hospital occupancy telemetry
Ambulance Agent (Mobile First Responder)	• Travel latency to incident scene • En-route vitals stabilization index • Turnaround cycle duration • Energy & fuel efficiency	• Dynamic urban traffic grid • Inclement weather conditions • Incident scene hazards • Hospital emergency bays	• Vehicle throttle, steering, brake • Emergency sirens & strobes • Bid proposal transmitter • Bed reservation request	• High-precision GPS receiver • Real-time traffic congestion API • Onboard vehicle diagnostics • Patient vitals (ECG, SpO2, BP)

📌 Mathematical Objective Formulation: Response Latency Weighted by Triage Urgency

$P_{dispatch} = \sum \gamma_i \cdot (t_{arrival} - t_{call}) + \text{Penalty}_{coverage}$. Parameter γ_i applies an exponential penalty multiplier for ESI-1 trauma vs ESI-5 routine cases, mathematically guaranteeing high-acuity priority.

PEAS: Hospital ER Agent & Clinical Triage Model

Rubric: PEAS (3 Marks)



Hospital Allocation Agent (PEAS)

- **Performance (P):** ER bed occupancy variance across city hospitals (load balancing), ambulance offload delay minimization (< 15 min target), specialized care match accuracy (Cath Lab, Trauma, Stroke), patient survival rate.
- **Environment (E):** Inbound ambulance flow, walk-in emergency department queue, ICU/CCU bed occupancy, operational surgical suites, doctor and nursing shift staffing.
- **Actuators (A):** Bed reservation confirm/reject signals, hospital diversion advisories (Code Yellow/Red), trauma resuscitation team paging, triage queue re-indexing.
- **Sensors (S):** Live Hospital Information System (HIS) bed telemetry, ambulance bay RFID vehicle sensors, incoming ambulance GPS countdown feeds, patient telemedicine streams.



Emergency Severity Index (ESI 1–5) Triage Hierarchy

ESI LEVEL 1: RESUSCITATION

Immediate life threat (Cardiac arrest, severe trauma). Immediate bed lock.

ESI LEVEL 2: EMERGENT

High risk, confused/lethargic, severe pain (Stroke, STEMI). Target < 10 mins.

ESI LEVEL 3: URGENT

Requires multiple resources, stable vitals (Moderate asthma, fractures). < 30 mins.

ESI LEVEL 4: LESS URGENT

Single resource required (Simple laceration, sprain). Target < 60 mins.

ESI LEVEL 5: NON-URGENT

No resources required (Medication refill, mild cold). Routine clinic care.

Environment Analysis across Russell & Norvig's 7 Dimensions

Rubric: Env & Agent (3M)

Canonical Dimension	EMS Environment Classification	Technical Justification & Algorithmic Impact
1. Observability	Partially Observable	Ambulance agents cannot observe distant traffic jams, internal patient bleeding, or sudden hospital bed status changes until sensor/telemetry sync.
2. Agent Multiplicity	Multi-Agent (Coop + Comp)	Ambulances cooperate toward city-wide life-saving metrics, but compete for shared resources: right-of-way on congested road lanes and limited emergency ICU beds.
3. Determinism	Stochastic	Call arrivals follow a non-homogeneous Poisson process; road congestion and patient health deterioration are non-deterministic, requiring probabilistic state models.
4. Episodic / Sequential	Sequential	Current dispatch choices directly exhaust ambulance availability in an urban quadrant, creating delayed ripple effects on subsequent incoming emergencies.
5. Static / Dynamic	Dynamic	Traffic speeds fluctuate, road blocks appear, and patient physiological states degrade continuously while agents deliberate routing and bidding decisions.
6. Discrete / Continuous	Continuous (Hybrid)	Spatial coordinates, velocities, and travel times are continuous; triage codes, hospital admission slots, and dispatch states are discrete.
7. Known / Unknown	Known Rules / Unknown States	The road graph topology, traffic physics, and medical triage protocols are known; only dynamic runtime states require continuous estimation.

Agent Typology & Internal Decision Architecture

Rubric: Env & Agent (3M)



Ambulance Agent

Utility-Based Agent

- Evaluates multi-attribute utility function:
 $U(a) = w_t \cdot (1/ETA) + w_s \cdot P_{survival} - w_c \cdot Cost$
- Trades off transit latency against destination hospital clinical capabilities (e.g., Cath Lab, Trauma Center).



Dispatcher Agent

Model-Based Goal Agent

- Maintains global world model tracking fleet distribution across city quadrants.
- Predicts coverage vulnerability voids.
- Balances immediate assignment against future readiness risks; triggers proactive relocation.



Hospital ER Agent

Model-Based Utility Agent

- Maintains queuing model of incoming ambulances vs internal ER throughput.
- Computes marginal cost of bed admission.
- Dynamically inflates admission bid cost as occupancy exceeds 85% to deter overcrowding.



Autonomous Agent Perception-Action Control Loop Architecture

1. SENSORS

GPS, Traffic API,
Patient Vitals

2. STATE ESTIMATOR

Belief update $b(s)$:
Congestion & Queues

3. UTILITY ENGINE

Multi-objective
Tradeoff Evaluation

4. ACTION SELECTOR

Optimal Policy $\pi(s)$
Decision Selection

5. ACTUATORS

Route, Bid, Bed
Reservation Lock

Multi-Agent Interaction & Contract Net Protocol (CNP)

Rubric: Env & Agent (3M)

PHASE 1: CFP ANNOUNCEMENT

- Dispatcher receives incoming 911 call.
- Categorizes ESI priority level (1–5).
- Broadcasts Call for Proposals (CFP) with GPS coordinates and clinical requirements to active units.

PHASE 2: AUTONOMOUS BIDDING

- Idle ambulances compute marginal cost:
$$\text{Bid}_i = \alpha \cdot T_{\text{travel}} + \beta \cdot T_{\text{service}} - \gamma \cdot \text{Readiness}$$
- Travel latency T_{travel} computed via Time-Dependent A* on live traffic.
- Submit proposals to Dispatcher.

PHASE 3: CONTRACT AWARD

- Dispatcher evaluates bids against global objective function.
- Awards contract to optimal utility bidder.
- Transmits rejection messages to secondary units to resume patrol.

PHASE 4: HOSPITAL BED LOCK

- Assigned ambulance conducts bilateral admission auction with candidate ER facilities.
- Locks trauma/ICU bed access prior to departure.
- Eliminates ambulance offload delays.

⚡ Protocol Resilience & Communication Standards

- **FIPA-ACL Standard:** Utilizes standardized performatives (CFP, PROPOSE, ACCEPT-PROPOSAL, REJECT-PROPOSAL, INFORM) for seamless agent interoperability.
- **Fault Tolerance:** If an assigned ambulance suffers mechanical failure, a localized re-bid is automatically triggered to the next best unit within 500 ms.
- **Computational Scalability:** Solves allocation locally in $O(k)$ parallel message passing rather than $O(n!)$ exponential centralized recalculations.

Algorithmic Modeling: Dynamic Search & Time-Dependent Routing

Rubric: Algo Search (3M)

Time-Dependent A* (TDA*) Algorithm

- **Road Graph Model:** City modeled as directed graph $G = (V, E)$. Edges have time-varying traversal costs $c(u, v, t)$ reflecting traffic conditions at arrival time t .
- **Cost Evaluation Function:** $f(n, t) = g(n, t) + h(n)$
 - $g(n, t)$: Actual travel latency from origin to node n at departure time t .
 - $h(n)$: Admissible heuristic estimate of remaining transit time to destination.
- **Admissibility Proof:** $h(n) = \text{Euclidean_Distance}(n, \text{Goal}) / v_{\text{max}}$. Because speed $v(e) \leq v_{\text{max}}$ on all road edges, $h(n) \leq h^*(n)$ holds universally, strictly guaranteeing optimal shortest-time paths without overestimation.
- **Time Complexity:** $O(|E| + |V| \log |V|)$ using priority queues. Pre-computed contraction hierarchies accelerate metropolitan queries to < 10 milliseconds.

Dynamic Re-planning Engine (D* Lite)

- **Congestion Trigger:** If live traffic sensors detect downstream edge delay inflation $\Delta t_{\text{edge}} > 25\%$, dynamic replanning is triggered.
- **Incremental Repair:** D* Lite reuses search trees from the previous A* execution, repairing only affected path segments in millisecond latency rather than re-computing from scratch.

Clinical Priority Escalation Re-route

- **Vitals Degradation:** If onboard medical sensors detect patient deterioration to ESI Level 1 en-route, destination recalculation is triggered to the nearest specialized trauma facility.
- **Green Corridor Preemption:** Ambulance agent transmits emergency V2X signal to traffic light controllers along the updated path, preempting red lights to clear congestion.

Hospital Allocation Optimization & Mathematical Modeling

Rubric: Algo Search (3M)

Multi-Criteria Global Optimization Function

$$\min J = \sum_i [w_1 \cdot T_{\text{resp}}(i, a) + w_2 \cdot T_{\text{transit}}(a, h) + w_3 \cdot T_{\text{handover}}(h) + w_4 \cdot \text{Penalty}(i, h)]$$

- $T_{\text{resp}}(i, a)$: Response duration from emergency call to ambulance scene arrival.
- $T_{\text{transit}}(a, h)$: En-route travel duration from scene to destination hospital h .
- $T_{\text{handover}}(h)$: Anticipated hospital offload delay based on current ER bay queue.
- $\text{Penalty}(i, h)$: Clinical mismatch penalty (Penalty = ∞ if hospital lacks required specialty such as Cath Lab or Burn Unit).
- **System Constraints:**
 1. $\sum_a y_{ia} = 1 \quad \forall i$ (Every emergency served by exactly one ambulance)
 2. $\sum_i x_{ih} \leq \text{Capacity}_h \quad \forall h$ (Hospital admissions $\leq$ available ER beds)
 3. $x_{ih}, y_{ia} \in \{0, 1\}$ (Binary decision variables)



Two-Sided Stable Matching (Gale-Shapley Adaptation)

Eliminates the 'herd effect' where ambulances blindly oversaturate the closest hospital.

Ambulance Preference Ranking Protocol:

$\text{Rank_amb}(h) = f(\text{Proximity}, \text{Predicted Offload Delay}, \text{Clinical Specialty Score})$
Ambulances rank hospitals that offer zero offload delay and certified care.

Hospital ER Preference Ranking Protocol:

$\text{Rank_hosp}(\text{amb}) = f(\text{Patient ESI Acuity}, \text{Estimated Arrival Time}, \text{Bed Readiness})$
Hospitals prioritize code-red resuscitation patients (ESI 1) over routine cases.

Stability Guarantee & DCOP Decentralization:

- Mathematically guarantees that no ambulance-hospital pair has a mutual incentive to break allocation, preventing oscillatory re-routing.
- Implemented via Distributed Constraint Optimization (DCOP) with linear message passing.

Project Execution Roadmap & Viva Defense Q&A

Rubric: Q&A & Plan (1M)



6-Week Implementation Plan Mapped to Review 2 Evaluation Rubric (10 Marks)

SPRINT 1 (WEEKS 1–2)

Tool/Package Setup (3M)

- Configure Python Mesa MAS simulation.
- Generate synthetic street graph via OSMnx.
- Implement Poisson emergency call generator.

SPRINT 2 (WEEKS 3–4)

Multi-Agent Execution (3M)

- Implement Contract Net Protocol messaging.
- Deploy Time-Dependent A* routing engine.
- Integrate bilateral hospital admission handshake.

SPRINT 3 (WEEKS 5–6)

Demo & Testing (4M)

- Stress test mass-casualty surge scenarios.
- Benchmark response time vs Centralized baseline.
- Code refactoring for horizontal scalability.



Viva Defense Q1: Why MAS over Centralized Linear Programming?

- Centralized solvers suffer from combinatorial state explosion $O(n!)$ during urban disasters, causing critical computation delays.
- Central dispatch represents a single point of failure.
- MAS decomposes global optimization into localized, parallel auctions with linear scalability and inherent fault tolerance.



Viva Defense Q2: How is hospital crowding & herd behavior prevented?

- Hospital agents dynamically inflate admission bid penalties as ER bed occupancy crosses 85% capacity.
- Two-sided stable matching naturally deflects non-critical ambulances to adjacent secondary facilities.
- Pre-arrival bed reservations guarantee zero offload delays upon ambulance arrival.